REVIEW

A new anchor point for gut microbiome to regulate complications of allogeneic hematopoietic stem cell transplantation: oxidative stress

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Allogeneic hematopoietic stem cell transplantation (allo-HSCT) is a cure for many malignant hematologic and metabolic diseases, whose complications are an unignorable cause of long-term prognosis. Pretreatment chemoradiotherapy administered before and after transplantation alters the state of the bone marrow hematopoietic microenvironment. Under the stress of hematopoietic injury, a large number of hematopoietic stem cells (HSCs) proliferate and differentiate, producing large amounts of reactive oxygen species (ROS). In addition, the chemotherapeutic pretreatment drugs also cause oxidative stress (OS) themselves, which can indirectly exacerbate OS of the bone marrow microenvironment. The bone marrow hematopoietic microenvironment and local OS affect the development of infections, relapse, graft-versus-host disease (GVHD), poor graft function (PGF), persistent thrombocytopenia (PT), chronic fatigue syndrome (CFS), bronchiolitis obliterans syndrome (BOS), and transplant-associated thrombotic microangiopathy (TA-TMA). Recent studies have demonstrated that the gut microbiome (GM) and its metabolites such as short-chain fatty acids (SCFAs), secondary bile acids, trimethylamine (TMA) and signaling molecules such as H_2S and NO can enter the bloodstream through the intestinal epithelium and reach all parts of the body. This review describes the impact of OS on the development of allo-HSCT complications. In this article, we will elucidate the mechanism of OS on allo-HSCT complications, summarize the interaction between GM and OS in the body, and highlight the role of GM regulation of OS in allo-HSCT complications to offer new strategies for the prevention and treatment of allo-HSCT complications.

HIGHLIGHTS

- Chemoradiotherapy pretransplant disrupts bone marrow, generating reactive oxygen species (ROS) from stressed hematopoietic stem cells (HSCs) and drugs, worsening oxidative stress (OS)
- High OS post-transplant exacerbates graft-versus-host disease (GVHD), infections, graft dysfunction, and thrombotic/immune disorders (e.g., transplant-associated thrombotic microangiopathy [TA-TMA] and bronchiolitis obliterans syndrome [BOS])
- Gut microbiome (GM) metabolites (short-chain fatty acids [SCFAs], bile acids) and signaling molecules (NO, H_2S) enter circulation, modulating OS and immunity
- Dysbiosis of the GM post-transplant may amplify systemic OS, indirectly worsening complications such as GVHD and infections. Conversely, GM-derived antioxidants (e.g., SCFAs) could mitigate OS and improve outcomes
- Targeting the GM to regulate OS offers novel strategies for preventing and treating allogeneic hematopoietic stem cell transplantation (allo-HSCT) complications, emphasizing interdisciplinary approaches for prognosis improvement

Allogeneic hematopoietic stem cell transplantation (allo-HSCT) is a combination of conventional therapy (chemotherapy, radiotherapy, or antibodies) and cell replacement therapy for the treatment of various benign and malignant hematologic diseases [1,2]. Allo-HSCT requires a preoperative procedure in which the recipient is given a dose of radiation or chemotherapy to clear the marrow and destroy all the tumor cells in the bone marrow to the maximum extent possible, followed by intravenous infusion of donor hematopoietic stem cells (HSCs) to re-establish the recipient's hematopoietic and immune function [3]. When killing tumor cells in the bone marrow, marrow

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cleaning therapy can also damage residual normal HSCs and progenitor cells, affecting long-term post-transplant hematopoietic reconstitution and hematopoietic progenitor cell (HPC) colony formation, promoting certain allo-HSCT complications such as graft-versus-host disease (GVHD) and poor graft function (PGF). Oxidative stress (OS) is involved in many of the complications of allo-HSCT. For example, increased plasma OS levels are related to the development of chronic graft-versus-host disease (cGVHD) [4]. OS levels in the hematopoietic microenvironment are associated with the development of PGF [5], and high levels of ROS-induced complement responses in the vascular microenvironment are closely linked to the aggravation of transplant-associated thrombotic microangiopathy (TA-TMA) [6].

Recent studies have shown that the gut microbiome (GM) plays an important physiologic role in human immunity, metabolism, and the pathogenesis of various diseases [7]. The signaling molecules (H_2S and NO) and metabolites (short-chain fatty acids [SCFAs], trimethylamine [TMA], and secondary bile salts) produced by microbial metabolism in the gut influence the OS balance of the gut and the body, regulating a range of biological functions [8,9]. Conversely, allo-HSCT affects the GM. For example, antibiotic treatment after allo-HSCT can cause increased levels of *enterococci, streptococci*, and various *Enterobacteriaceae* species, which are otherwise in low abundance in the GM, thereby altering the structure of GM components [10]. GM interactions with allo-HSCT organisms can affect levels of OS in the bone marrow hematopoietic microenvironment.

This article reviews the effect of GM on allo-HSCT complications, demonstrates a direct link between OS and the formation and development of allo-HSCT complications, suggests that the effect of GM on allo-HSCT complications may be related to GM affecting the level of OS in the body, with a view to providing new ideas for the clinical prevention and treatment of allo-HSCT complications.

OS AFFECTS ALLO-HSCT COMPLICATIONS

OS Affects Hematopoietic and Immune Reconstitution

OS refers to a state of altered redox balance in the body with increased levels of free radicals or some reactive metabolites, mainly including ROS and reactive nitrogen species (RNS) [11]. The accumulation of ROS leads to intracellular DNA damage, gene expression alterations, and changes in the activity of key metabolites. It also determines cell survival or death and is associated with hematopoietic recovery [12]. Manipulations such as radiation, growth factor stimulation, or chemotherapy involved in transplantation cause OS in the bone marrow microenvironment, with large numbers of HSCs entering the cell cycle to expand massively from quiescence and generating large amounts of ROS. The excessive accumulation of ROS is detrimental to the recovery of hematopoietic and immune function after transplantation [13].

OS affects hematopoietic function mainly by affecting HSC function and microenvironmental matrix components. The hematopoietic factors interleukin (IL)-3 and erythropoietin (Epo) within the bone marrow hematopoietic microenvironment [14], the expression of Akt protein [15], and the absence of tuberous sclerosis complex 1 (TSC1) [16] promote ROS production, and the accumulation of ROS can activate various kinases and phosphatases to enter the cell cycle through a redox-related signaling cascade in HSCs [13]. Activation of the PI3K/Akt/mTOR signaling pathway stabilizes c-Myc, induces G1 phase cell cycle protein p27 expression, promotes hematopoietic stem and progenitor cells (HSPCs) to progress from the G1 phase to S phase, and accelerates cell cycle progression [17]. FOXO-type transcription factors regulate ROS levels to inhibit the PI3K/AKT pathway, maintain the DNA repair response in stem cells, and maintain normal HSC function [18]. Under stress, the S1P/S1P1 Axis [19] and the CXCL12/CXCL4 Axis are activated to induce elevated ROS levels in HSCs through the HGF/c-Met pathway [20,21], driving the migration of hematopoietic cells. DNA-binding protein 2 (ID2) plays a role in protecting the HSC genome under stress, and HIF-1α positively promotes ID2 expression to enhance protection [22]. In addition, chronic inflammation due to sustained OS can induce inflammation-activate peroxisome proliferator-activated receptor γ (PPAR-γ) transcription and promote HSC senescence [23]. The effects of ROS may not all be detrimental, with some beneficial effects noted. Recent studies have revealed that ROS may play a favorable role in HSCs through the HIF-YAP-Notch signaling pathway, but the specific functions of HSCs are still being explored [24,25].

Microenvironmental matrix components also play an important role in hematopoietic reconstitution and may protect HSPC function by regulating ROS. Cellular experiments have shown that human mesenchymal stromal cells (MSCs) are able to scavenge ROS and RNS and effectively reduce OS damage [26]. The gap junction protein Connexin-43 (Cx43) expressed by HSPCs can transfer intracellular ROS to stromal cells to protect HSPCs and facilitate hematopoietic reconstitution [27]. Low ROS levels in MSCs are beneficial in stabilizing p53R172P, protecting MSCs from apoptosis, and facilitating long-term differentiation and proliferation [28]. On the contrary, high ROS levels halt the MSCs cell cycle and induce apoptosis [13]. However, lower ROS levels are not necessarily better because ROS levels that are too low tend to promote the differentiation of immature stromal cells to osteoblasts [29].

Furthermore, OS can affect the re-establishment of immune function after transplantation. Innate immune cells produce large amounts of ROS to kill pathogenic microorganisms, and the accumulation of ROS causes OS, leading to endothelial dysfunction. To ensure vascular tone and structure, effector T lymphocytes are activated to differentiate into Th-1, Th-17, and T regulatory (Treg) cells to regulate vasoconstriction and diastole, and the secondary transfer of Tregs reduces OS in the vasculature [30]. Ataxia-telangiectasia mutated (Atm) is associated with T-cell proliferation, and Atm decreases the ROS produced by T-cells to promote T-cell proliferation [31]. Excessive ROS kills activated CD4(+) T-cells and plasmacytoid dendritic cells (pDCs) affecting recipient immune function (Fig. 1) [32].

Poor Hematopoietic Reconstitution and Allo-HSCT Complications

In the OS state, excess ROS activates multiple pathways that impair HSPC function and the balance of the hematopoietic microenvironment, causing impaired hematopoietic reconstitution. Delayed or incomplete hematopoietic recovery after transplantation can lead to poor graft function (PGF), a common complication after allo-HSCT. PGF is often combined with infection and bleeding and requires urgent secondary transplantation [33], including increased levels of ROS in the bone marrow hematopoietic microenvironment [34], post-transplant immune dysregulation-mediated hematopoietic breakdowns, donor allozygous anti-human leukocyte antigen (HLA) antibody-mediated immune rejection, cytomegalovirus (CMV)

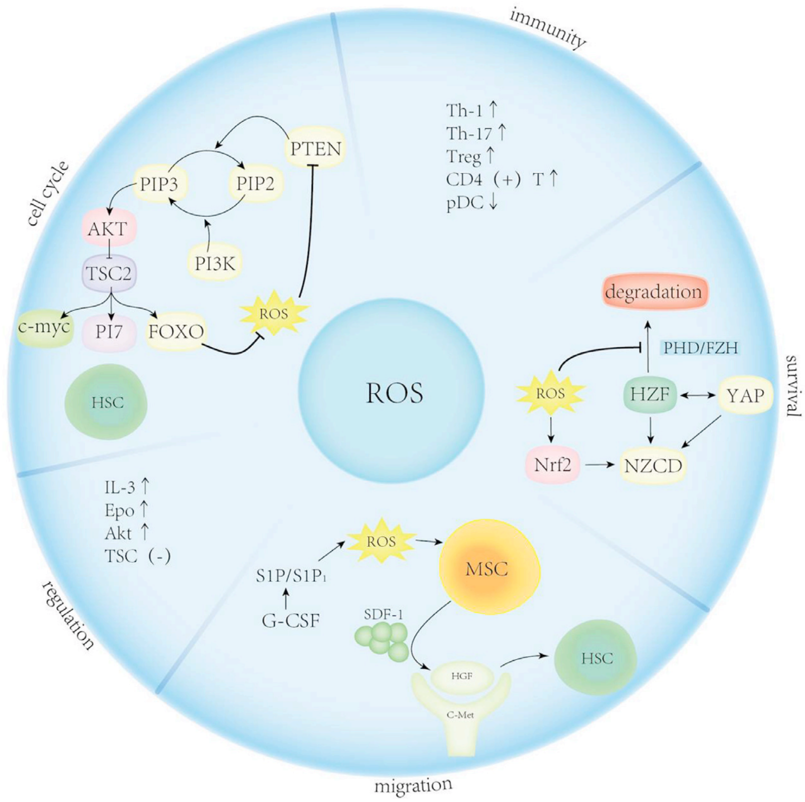

Figure 1 The Biological effect of ROS in bone marrow environment. ROS are involved in multiple signaling pathways and regulate immunity in bone marrow microenvironment as well as cell cycle, migration, and degradation of HSCs. ROS is regulated by several factors. IL-3, Epo, and Akt protein expression as well TSC1 deficiency promotes ROS production. ROS activates the PI3K/Akt/mTOR signaling pathway, stabilizes c-Myc, and induces the G1 phase of the cell cycle and protein p27 expression to accelerate cell cycle progression. This can be inhibited by FOXO. ROS activates Nrf2 to trigger the degradation of HSCs, and facilitate HSC survival via the HIF-YAP-Notch signaling pathway. In addition, elevated G-CSF levels trigger S1P/S1P1 and increase ROS, which stimulate MSCs to secrete SDF-1. In addition, it activates the HGF/c-Met pathway to promote HSC migration. Finally, ROS can increase CD4 + cells, including Th-1, Th-17, and Treg, but decreases pDCs.

infection [35], and iron overload. Among these, high ROS in the bone marrow hematopoietic microenvironment is highly correlated with the development of PGF, which is highly statistically significant. The mechanism is that ROS in the bone marrow microenvironment leads to a substantial decrease in CD34-positive cells [36]. A decrease in mesenchymal stem cells [37] and endothelial cells (ECs) [38] triggers damage to the bone marrow endothelial and vascular microenvironment [39], leading to the development of PGF. Prophylactic intervention with antioxidant drugs such as N-acetyl-l-cysteine (NAC) may prevent the development of PGF by reducing ROS levels and may repair ECs in the damaged bone marrow hematopoietic microenvironment to accelerate hematopoietic reconstitution [40].

Persistent thrombocytopenia (PT) manifests as recovery of other blood cell lines after allo-HSCT. In addition, consistently low platelet counts are observed [41] primarily due to hypersplenism or autoantibody-induced immune platelet destruction [42]. High ROS levels in the bone marrow hematopoietic microenvironment can further aggravate the development of PT by modulating the integrity of the ECs of the hematopoietic microenvironment as well as inflammation, affecting the differentiation of megakaryocytes and consequently platelet production [43].

OS Affects GVHD

The inflammatory response due to ROS accumulation exacerbates damage in both aGVHD and cGVHD target cells [44]. This is because elevated endogenous mitochondrial ROS induces the nuclear factor (NF)-κB pathway and activates NLRP3 inflammation-related signaling to induce macrophage polarization as well as Th1 and Th17 differentiation, exacerbating inflammatory infiltration and triggering a stronger T-cell immune response to GVHD target cell [45]. In addition, ROS can activate high mobility group box 1 (HMGB1) signaling molecules related to innate immunity and promote the development of GVHD [46]. Reduced scavenging of ROS by indoleamine 2,3-dioxygenase 1 (IDO1) is noted in the bone marrow hematopoietic microenvironment [47]. Reduced migratory capacity of bone marrow endothelial progenitor cells and increased ROS levels after transplantation are associated with GVHD pathogenesis [48]. Chronic inflammation of the central nervous system (CNS) is one of the important pathologic features of cGVHD and is associated with donor bone marrow sources expressing major macrophage infiltration of histocompatibility complex (MHC) II [49] Inhibition of the JAK/STAT pathway reduces ROS levels attenuating interferon (IFN)-γ overexpression and reduces inflammation modulating donor macrophage MHC II expression, thereby improving CNS cGVHD outcomes [50].

GVHD is often observed in the skin, gut, liver and lung of patients [51]. OS at the site of the lesion also affects GVHD. In the lung and liver, keratinocyte growth factor prevents and attenuates GVHD-related organ damage by elevating glutathione (GSH) levels and decreasing ROS levels [52]. Human placenta-derived mesenchymal stromal cells (hPMSCs) can reduce inflammation in GVHD mice by increasing GSH and glutathione S-transferase (GST) levels and downregulating ROS production by modulating the interaction of nuclear factor erythroid 2-related factor 2 (Nrf2) and NF-κB signaling pathways, decreasing programmed cell death (PD)-1 expression on donor CD4 T-cells, and reducing inflammation accompanied by improved histopathologic changes in the liver [53,54]. Reducing the expression levels of NFκB and T-bet reduces ROS and the amount of donor T-cells in GVHD target organs and attenuates GVHD [55]. cGVHD is typically characterized by intestinal damage. Inflammatory bowel disease (IBD) is often associated with increased myeloperoxidase (MPO) activity and increased production of inflammatory cytokines [56], and OS exacerbation can promote increased intestinal symptoms in cGVHD. For example, arsenic trioxide (ATO) can deplete glutathione and promote ROS production, killing activated donor CD4 (+) T-cells and pDCs and reducing skin fibrosis in scleroderma cGVHD mice [57].

OS and Other Allo-HSCT Complications

OS has also been associated with other allo-HSCT complications. Elevated systemic iron levels, ROS levels [58], and elevated intracellular ROS levels in T-cells [59] increase the risk of infection. Widespread vitamin C deficiency and diminished antioxidant capacity could be seen in patients undergoing HSCT, with higher OS in vivo, which leads to reduced neutrophils and greater susceptibility to infection [60]. The reduction in neutrophils often causes FN, and increased OS is associated with persistent attacks of febrile neutropenia (FN) [61].

TA-TMA is one of the fatal complications after allo-HSCT, with a morbidity of 0%−64% and mortality of 60%−75% [62]. TA-TMA manifests as complement-activated endothelial damage [63]. Recent studies have found that damage to the vascular endothelium is associated with OS and that elevated ROS leads to increased apoptosis of ECs [6]. Modulation of ROS by heme oxygenase 1 (HO-1) [64] and the Nrf2 pathway [6] inhibits activation of the TA-TMA-associated complement pathway, thereby protecting ECs from OS injury.

Elevated ferritin concentrations, NO levels, and neutrophil counts in the lungs due to transplantation exacerbate OS in the lungs, elevating ROS levels, leading to airway injury or increased permeability of microvessels, and triggering or exacerbating BOS [65]. The onset and progression of CFS is associated with increased ROS levels due to mitochondrial dysfunction. In addition, exposure to ionizing radiation in transplantation is a predisposing factor [66]. Mitochondrial dysfunction occurs in leukocytes and muscle cells, and patient experiences muscle pain. However, in CNS cells, symptoms occur as systemic hypersensitivity and chronic widespread pain [67].

OS and Iron Overload

Iron is an important factor affecting OS levels and can cause toxicity by promoting ROS production, inducing inflammation, and contributing to tissue damage and organ dysfunction. Iron overload is relatively common in HSCT recipients, with a 30%−60% probability of occurrence [68], which often due to repeated transfusions before and after transplantation, increased iron uptake due to intestinal inflammation caused by chemotherapy, disruption of iron homeostasis triggered by bone marrow ablation, and increased iron release from damaged tissues [69]. Excess iron causes OS, and excess ROS activates multiple pathways that impair HSPC function as well as hematopoietic ecologic niches, causing impaired hematopoietic reconstitution. In addition, excess iron is associated with a variety of short- and long-term allo-HSCT complications, such as infection, enterocolitis, liver dysfunction, and GVHD [69−71].

Serum iron levels are important predictors of prognosis in allo-HSCT. A retrospective analysis of 250 patients undergoing allo-HSCT found a significant association between increased pretransplant serum ferritin (SF) levels and reduced overall survival and increased transplant-related mortality [72]. A recent meta-analysis showed that SF levels were associated with predicting complications of allo-HSCT and that elevated pretransplant SF levels were associated with a higher incidence of infection and a lower incidence of cGVHD, but not acute graft-versus-host disease (aGVHD) [73].

GM INFLUENCES THE DEVELOPMENT OF ALLO-HSCT COMPLICATIONS

GM Affects Hematopoietic Reconstitution in Allo-HSCT

The chemotherapy or total body irradiation (TBI) involved in allo-HSCT produces "bystander effects" on nontargeted cells. It damages transplanted HSC, destroys sinusoidal vessels, and depletes the MSCs, causing damage to the myelopoietic microenvironment and myelopoietic capacity [74]. The recovery of hematopoietic function after allo-HSCT is the key to positive prognosis.

The GM itself and its metabolites affect the re-establishment of bone marrow hematopoiesis after transplantation. The GM directly promotes bone marrow progenitor cell differentiation through MyD88/TICAM signaling and toll-like receptor (TLR) signaling [75]. The GM induces MSCs to produce cytokines through NOD1 ligand −NOD1 signaling [76], ensuring proper hematopoiesis. Moreover, the GM promotes HPC differentiation, which is associated with anti-infective action and enhances immunity against pathogens by promoting the differentiation of bone marrow progenitor cells to granulocytes or monocyte progenitor (GMP) cells [77]. The GM affects the interaction of granulocytes and ECs to increase ROS or certain cytokines, stimulating the differentiation of HPCs to bone marrow granulocytes [78]. Post-transplant antibiotic therapy often causes GM disorders accompanied by hematopoietic suppression of multiple HPC sublines [79], which decreases granulocyte production, increases susceptibility to sepsis and septicemia [80], and promotes FN [61].

The GM also affects platelet reactivity, and this is related with the development of immune thrombocytopenia (ITP). In addition, GM metabolites can agonize platelet activation, leading to an increased risk of thrombosis [81]. In patients with ITP, there is a tendency for an increase in *Bacillus mimicus* and a decrease in thick-walled bacteria in the intestine. GM dysregulation causes a decrease in platelet count in patients with ITP, but platelet function appears to be enhanced in a compensatory manner [82]. The depletion of secondary bile acids affects the immune system [83]. All of these effects promote the onset of ITP. Additionally, intestinal bacteria and metabolites that change with the onset of ITP can be used as biomarkers for the diagnosis of ITP. Specifically, increases in *Vibrio* and *Streptococcus acnes*; increased Cer (t18:0/16:0), Cer (d18:1/17:0), and 13-hydroxyoctadecanoic acid levels; and elevations in the fecal metabolites fatty acyl and glycerophospholipids serve as the basis for the diagnosis of ITP [84].

The GM Affects Immune Reconstitution of Allo-HSCT

Radiation and chemotherapy administered to patients undergoing allo-HSCT interfere immune system severely and cause impairment of thymic function, causing a significant decrease in the number of CD4+ T-cells, neutrophils, and leukocytes [85]. Donor stem cells differentiate into leukocytes that are released into the bloodstream, restoring leukocyte counts and initiating immune reconstitution. Longitudinal analysis of circulating immune cell counts and microbiota samples from hundreds of patients undergoing allo-HSCT revealed a concordance between intestinal bacteria and immune cell kinetics, with an abundant GM associated with faster immune recovery [86].

The restoration of immune function involving T-cells is essential to prevent complications and prolong survival [87]. The GM regulates immune homeostasis, which also plays an important role in immune reconstitution after transplantation [88]. The GM can directly regulate the differentiation and proliferation of immune cells [89]. It was found that segmented filamentous bacteria, *Akkermansia muciniphila* and *Clostridium*, activate Th17 and Treg [90,91]. *Clostridium* secretes SCFAs and polysaccharide A inducing the differentiation of T-cells into Treg cells [92]. Th17-mediated intrinsic immunity defends against pathogenic microorganisms in the intestine, and Treg cells suppress the proinflammatory effects of high cytokine stimulation of Th17, effectively strengthening intestinal immune function.

The GM can regulate the developmental differentiation of innate immune cells by promoting hematopoiesis through TLR signaling in HSCs or overall digital immune reconstitution in a homograft model [77]. Promoting hematopoiesis to regulate the developmental differentiation of innate immune cells. The GM enhances the differentiation of bone marrow progenitor cells to GMP and facilitates anti-infection effects [77]. Conversely, the GM can also affect hematopoiesis by influencing the differentiation of immune cells. A healthy GM promotes Th17 cell differentiation and expansion, triggering autoimmune-mediated acquired bone marrow failure syndrome [93]. Mice with GM disorders caused by antibiotics exhibit delayed immune reconstitution [79]. Poor recovery of the hematopoietic and immune systems after transplantation is associated with the occurrence of complications such as infection [94], PGF, PT, and even relapse. Promoting immune recovery after transplantation facilitates prolonged post-transplant recovery. For example, results of T-cell depleted graft transplantation, a therapy that promotes recovery of T-cell function, from 554 pediatric and adult patients who received allo-HSCT showed a reduced chance of relapse and prolonged post-transplant survival [95].

The GM Affects the Risk of Secondary Infection in Patients Undergoing Allo-HSCT

A healthy GM is mainly composed of thick-walled bacteria and the phylum Bacteroides, as well as *Aspergillus, Actinobacter, Verruciformes,* and *Clostridium* [96]. The GM plays an critical part in maintaining the integrity of the intestinal barrier and regulating immune cell differentiation and cytokine levels, maintaining inflammatory homeostasis in vivo to keep the body at a low level of inflammation. In response to pretreatment before allo-HSCT, antibiotic therapy [97] and alterations in intestinal osmolality [98] can cause changes in the composition of the GM and secondary GM disorders. Disruption of the GM significantly affects the normal physiologic function of mucosal and normal physiologic function of immune cells throughout the body, and the host immune system in turn influences the composition of the GM. This forms a feedback loop [99], increasing the risk of infection in patients undergoing allo-HSCT.

Intestinal barrier integrity is a critical component of intestinal immunity, and the GM can enhance the infection chance in patients undergoing allo-HSCT by impairing intestinal barrier integrity. Intestinal barrier damage caused by high-intensity chemotherapy in allo-HSCT pretreatment is associated with early *Clostridium difficile* infection in allo-HSCT [100]. Intestinal mucosal chemical defense barrier composed of antimicrobial peptides (AMPs) secreted by Paneth cells and intestinal epithelial cells maintains inflammatory homeostasis through antimicrobial responses [101]. TLRs on the surface of intestinal microbes mediate the expression of AMPs regenerated islet-derived protein III-γ (RegIIIγ) [102,103]. As GM metabolites, SCFAs exert numerous effects:

1. SCFAs promote plasma cell-mediated secretion of IgA to enhance intestinal immune function [104,105].
2. SCFAs act on the GPR43 receptor to activate mTOR and STAT3 pathways in Th1 cells, thereby upregulating the expression of transcription factor Blimp-1 and promoting the secretion of IL-10 by Th1 cells to suppress inflammation [106].
3. SCFAs activate G protein-coupled receptors, inhibit histone deacetylase expression, increase the mRNA and protein expression of factors associated with the intestinal tight junction, and reduce intestinal permeability to ensure the integrity of the intestinal barrier.

In addition, tryptophan metabolites also regulate intestinal barrier function through aromatic hydrocarbon receptors [107]. The integrity of the intestinal mucosa is related to GM homeostasis.

The damage to the mucosal barrier and the shift in intestinal dominance caused by allo-HSCT can greatly increase the risk of microbial translocation infections and the pathogenicity of opportunistic pathogens. Strain analysis of blood from allo-HSCT recipients revealed that *E. coli* and *Klebsiella pneumoniae* strains that cause sepsis may originate from the intestinal tract [108]. Taur et al [109] studied 94 allo-HSCT recipients and reported a ninefold increased risk of bacteremia in vancomycin-resistant enterococci or gram-negative bacteria, respectively, with the predominance of enterococci or saprophytes. Fecal microbial transplantation (FMT) can be used to directly alter the recipient's intestinal microbiota to normalize its composition and obtain therapeutic benefits, and current clinical applications have shown FMT to be effective in the prevention and treatment of *C. difficile* infections [110]. Metabolites of the GM also play an influential role in infections. The microbial metabolite butyrate enhances intestinal barrier function and mucosal immunity to reduce the risk of infection [111]. The metabolite indole-3-aldehyde produced by the interaction of intestinal bacteria with tryptophan activates the aryl hydrocarbon receptor on innate lymphoid cells and induces the secretion of IL-22. IL-22 stimulates Paneth cells to secrete RegIIIγ, which maintains flora balance and acts as a defense against pathogenic bacterial infections

[112]. The metabolite secondary bile acids of *C. difficile* can defend against *C. difficile* infection [113].

Furthermore, GM is associated with the antiviral capacity of the organism. The outer membrane (OM)-associated polysaccharide A of enterobacteria can induce colonic dendritic cells secret IFN-β via TLR4−TRIF signaling and regulate IFN-I responses through IFN-β regulation of downstream interferon-stimulated gene expression to enhance the body's antiviral capacity [114]. The GM also affects the risk of viral infection after transplantation. A macrogenomic analysis of the gut virome in patients with intestinal GVHD found increased persistent of DNA viruses (anaplasma, herpesvirus, oncolytic virus, and polyomavirus) and decreased phage abundance. In addition, Picobina virus predicted the likelihood of severe intestinal GVHD [115]. The gut microbial metabolite butyrate reduces the incidence of lower respiratory viral infections in allo-HSCT recipients [116].

GM Influences GVHD

GVHD is also closely associated with GM. Lipopolysaccharides on the surface of bacteria can cross the intestinal mucosal barrier to activate the innate immune system, further activating donor T-cells and triggering GVHD [99,117]. Clinical studies have found differences in microbial ecology and metabolism in the gut between patients who develop GVHD after hematopoietic cell transplantation (HCT) and those who do not. This was demonstrated by increases in the abundance of *Clostridia, Porphyromonas*, and *Sharpella* in the gut of patients with GVHD, active short-chain fatty acid metabolism, β-oxidation and acyl-coenzyme A synthesis, and γ-aminobutyric acid shunting. Reduced GM abundance after allo-HSCT, especially in *C. perfringens* (*Blautia*) spp. and *A. muciniphila,* showed a strong correlation with the progression aGVHD [118]. Increased abundance of GM intestinal genes and clostridial *Blautia spp.* often means lower GVHD mortality and higher survival [119]. *Trichoderma* spp. and *Clostridium* spp. of the Akkermansia family (butyrate as the main metabolite) and low levels of *E. faecalis* abundance are associated with the development of chronic GVHD [120]. In contrast, the expansion of enterococci exacerbates intestinal inflammation and GVHD [121].

GM metabolites have important implications and predictive significance for the development of GVHD. Analysis of data from patients with cGVHD revealed significantly lower concentrations of butyrate and propionate in the intestine and circulation. High circulating concentrations of SCFAs were associated with a low incidence of cGVHD [120]. Decreased butyrate levels are associated with increased risk of GVHD. An evaluation system based on CoA-transferase acetate (BCoAT), a key enzyme for butyrate synthesis, is an important predictor of high risk intestinal graft-host disease [122]. The metabolism of 3-indolyl sulfate (3-indoxyl), produced by the genera *Thicketidae, Trichoderma* spp. and *Rumenococcaceae*, acts as an anti-inflammatory agent and contributes to intestinal ecologic homeostasis. Therefore, 3-indoxyl is an important biomarker for assessing intestinal health and predicting GI-GVHD risk and survival [112,123].

The Effect of GM on Other Allo-HSCT Complications

The abundance of each microbial community and metabolites of the GM is also associated with the development of other allo-HSCT complications. Post-transplant hematologic tumor recurrence is another important reason for allo-HSCT failure. The GM participates in the regulation of the body's immune response and plays an important role in tumor immunosurveillance, which can affect the risk of post-transplant tumor recurrence in patients undergoing allo-HSCT [50−52]. A retrospective analysis of 541 patients who underwent allo-HSCT for malignant hematologic diseases found that an increased abundance of *Eubacterium limosum* was negatively associated with relapse [53].

The development of CFS, also known as myalgic encephalomyelitis (ME), can be caused by an upregulation of systemic proinflammatory cytokine levels secondary to allo-HSCT. It was found that the abundance of intestinal *Selenomonas* spp., *Streptococcus alginolyticus* spp., and proinflammatory *Aspergillus* spp. and *Rhodobacter* spp. was significantly higher in allo-HSCT patients with CFS than those without CFS, and the mechanism may be related to the fact that the proinflammatory environment of the gastrointestinal tract may further promote the translocation of microorganisms, exacerbate the systemic inflammatory response, and aggravate the fatigue symptoms [44]. Microbiome sequence datasets can be used to predict disease status due to the unique GM presentation of subjects suffering from CFS [54].

BOS, also called pulmonary cGVHD, is triggered by an attack on the lungs by the donor immune system and is one of the lethal complications of allo-HSCT [124]. Changes in microbiota often precede clinical manifestations in BOS. Microbiomic studies have found that the microbial composition of the lung affects lung disease mainly by influencing inflammation and immunity and can predict the occurrence and prognosis of BOS. In addition, high levels of gram-positive bacteria can reduce BOS susceptibility, implying a better prognosis [125]. Chronic bacterial airway infections increase susceptibility to BOS, leading to a worse prognosis [126]. Pulmonary *Pseudomonas aeruginosa* (Pa) and nontuberculous mycobacteria (NTM) are stationed in the gut, with possible microbiota exchange through potential reflux symptoms and misaspiration in patients, thus causing infections [127]. A significant decrease in *Paramecium* spp. bacteria in the gut predicts chronic Pa colonization in the lung [128], whereas Pa upregulates glutamate-leucine-arginine-positive (ELR(+)) CXC chemokines, exacerbates BOS, and increases the risk of death [129]. This effect of the GM on lung microorganisms may be related to the BOS process.

GM INFLUENCES OS IN ALLO-HSCT

The GM Regulates Local OS

The GM is involved in the regulation of OS mainly through the regulation of metabolite (mainly SCFAs, TMA, bile acids), H_2S, and NO levels [130]. SCFAs play a major regulatory role among GM metabolites [131]. Butyrate in the intestinal lumen activates peroxisome proliferator-activated receptor (PPAR)-γ and promotes mitochondrial β-oxidation as well as oxidative phosphorylation in colonocytes. Acetate and butyrate in the blood can inhibit OS in many cells, such as islets and β-cells, enhancing cell viability and prolonging cell life [132]. In addition to SCFAs, TMA in the gut enters the bloodstream where it is synthesized in the liver as trimethylamine-N-oxide (TMAO), which affects OS by altering lipid metabolism, leading to platelet overactivity, obesity and cardiovascular disease (CVD) [8]. Bile acids stimulate mitochondrial production of ROS, promoting their release from neutrophils and macrophages causing oxidative damage to tissues and inducing the formation of several vasoactive mediators such as endothelin-1, cysteinyl leukotrienes, and F2-isoprostane [133] that impair renal function and exacerbate renal failure [134]. In contrast, bile acid levels are influenced by intestinal *Lactobacillus,*

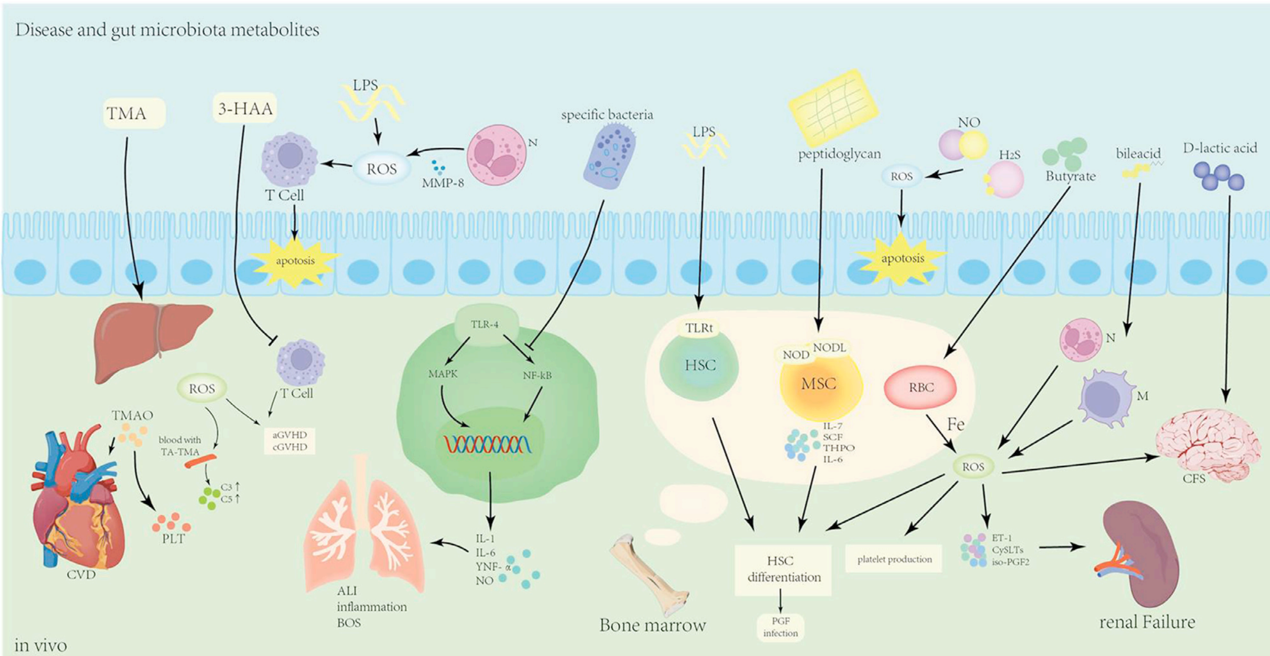

Figure 2 Various GM metabolites affect various physiologic activities of the body by regulating ROS. Gut microbiome metabolites regulate ROS levels in the intestine by inducing apoptosis of intestinal epithelial cells and disrupting intestinal integrity. Besides, metabolites in the intestine can enter the circulatory system through the intestinal wall to different parts of the body and induce or aggravate post-HSCT complications by regulating ROS levels.

Bifidobacterium, *C. perfringens*, and *B. mimicus* secrete bile salt hydrolases [135−137].

H_2S is mainly derived from the cysteine metabolism of the GM [138]. H_2S is highly efficient in scavenging ROS and RNS and exerts a long-lasting antioxidant effect [139]. However, high concentrations of H_2S inhibit mitochondrial cytochrome oxidase, promote ROS production, and induce mitochondrial depolarization and apoptosis, causing adverse effects [140]. Nitric oxide (NO), a by-product of the antioxidant process of H_2S [141], has a dual effect on pathogenic bacteria and OS. NO inhibits the adhesion of NO-sensitive bacteria enterohaemorrhagic *E. coli* (EHEC) to intestinal epithelial cells, effectively inhibiting microbial proliferation and preventing infection [142,143]. However, it facilitates the proliferation of the NO-tolerant bacteria *Salmonella* and increases the risk of infection [144]. Normally NO scavenges oxygen radicals [145], but excessive levels of NO molecules react with superoxide ions to generate peroxynitrite, leading to increased levels of ROS/RNS, increasing redox potential, inducing apoptosis in enterocytes, inhibiting proliferation and migration of intestinal epithelial cells, and promoting inflammatory responses [146,147].

GM regulation of local levels of OS affects allo-HSCT complications. In fact, the severity of human intestinal GVHD depends on the infiltration of neutrophils at the location of the lesion, and the production of chemokines induced by matrix metalloproteinase (MMP)-8 and catabolic lipopolysaccharide (LPS) produced by neutrophils [148] elevates ROS levels to activate T-cells [149] and exacerbate tissue damage. Schwab et al. [150] found that local GM determined the level of neutrophil infiltration and that exacerbation of GVHD in the intestine was achieved by GM through neutrophil regulation of ROS levels. Moreover, 3-hydroxyanthranilic acid (3-HAA), produced by tryptophan catabolism in response to the GM, reduces GSH levels in activated T-cells, enhances OS, induces apoptosis, selectively eliminates activated T-cells, and attenuates aGVHD [151]. The GM affects the CNS via the gut−brain axis involving neurologic and metabolic pathways [152]. For example, the GM can stimulate the production of neurotransmitters such as dopamine and signaling molecules such as gamma-aminobutyric acid and neurotoxic metabolites to affect post-traumatic stress [153]. The pathogenesis of CFS is more supported by the D-lactate theory, which is now generally supported as a result of bacterial fermentation and accumulation of D-lactate leading to its increase in the blood, which then affects brain regions causing the corresponding neurologic symptoms [154]. The antibiotic treatment required for transplantation also contributes to the development of CFS, with antibiotics targeting mitochondria to produce higher levels of ROS causing oxidative damage and exacerbating CFS [155]. Acute lung injury (ALI) is one of the common complications after transplantation [156], and GM imbalance modulates the TLR4/NF-kB signaling pathway in the lung immune system, activating OS in the lung and mediating ALI through modulation of the intestinal barrier (Fig. 2) [157].

The GM Affects OS in the Hematopoietic Microenvironment

Changes in GM and intestinal wall permeability can cause changes in cytokine levels, inflammatory responses, and OS levels within the myelopoietic microenvironment via intestinal−marrow axis transduction, further affecting HSC function [158]. Kovtonyuk et al [159] experimentally demonstrated that alterations in the composition of the gut microbiota and the release of bacterial lipopolysaccharides and microbial-associated TLR ligands into the blood increased levels of the proinflammatory cytokines IL-1a and b (IL-1a/b) in the bone marrow hematopoietic microenvironment, skewed HSC differentiation toward the myeloid lineage. and promoted HSC senescence. The imbalance of GM promotes the differentiation and expansion of Th17 cells, which subsequently migrate to peripheral lymphoid tissues to act against pathogenic cytotoxic T-cells (CTLs) produced against specific antigens on HSCs. These cells cooperate with myeloid suppressor cytokines to induce apoptosis of HSC cells and ultimately lead to the development of acquired bone marrow failure syndrome [93].

GM-derived propionate promotes HSC differentiation toward megakaryocytes and promotes platelet formation and functional maturation through histone 3 acetylation and propionylation [160]. In a study of benzene exposure-induced hematopoietic damage, it was found that the hematopoietic damage caused by benzene inflammation could be attenuated by adjusting the abundance of Family_XIII_AD3011_group and *Anaero truncus* sp. The mechanism is that adipic acid and 4-hydroxyproline produced by Family_XIII_AD3011_group and *A. truncus* sp. in GM enter the bone marrow microenvironment and activate TH2 cells to secrete IL-5 in large quantities, causing a proinflammatory or immune response [161]. The promarrow lineage differentiation of HSCs in obese mice can be reversed by antibiotic therapy or fecal transplantation, which may be explained by a decrease in Bacteroidetes and Firmicutes in the GM of obese mice and an increase in the levels of Verrucomicrobia, Actinobacteria, and Proteobacteria in the cecum and ileum, the metabolites of which are involved in the regulation of PPARγ2, Jag-1, SDF-1, and IL-7 genes. These genes subsequently affect the bone marrow microenvironment [162]. Lee et al. [163] found that microbiota-derived lactate induced SCF secretion by leptin receptor-expressing (LepR) MSCs through activation of Gpr81, promoting hematopoiesis and erythropoiesis. The GM has also been implicated in the prevention or restoration of humoral immune damage caused by chemotherapy. Approximately 40% of PCs in the bone marrow secrete IgA, of which approximately 40% express β7-integrin and CCR10, suggesting a mucosal origin [164]. The conserved antigen urea protein lipoprotein (MLP) on *E. coli* and *Salmonella* in the intestine is capable of inducing a systemic IgG response [165]. Moreover, MPL can be recognized by human antisera [166] and can have an impact on the human immune system [167].

Recent studies have found that the GM can regulate the recirculation of red blood cells (RBCs) by macrophages in the myelopoietic microenvironment under stressful conditions, modulate the level of iron in the myelopoietic microenvironment to influence HSC self-renewal and differentiation, and promote hematopoietic regeneration. The effect of GM on the myelopoietic microenvironment may occur through the metabolite SCFAs, most likely butyrate, which regulates RBC phagocytosis in myelopoietic microenvironment macrophages and degradation of iron redistribution through gene expression [161]. The redistribution of iron by the GM within the myelopoietic microenvironment affects iron levels and also OS. The GM can further influence the severity of associated allo-HSCT complications such as PGF [39], TA-TMA [6], PT [43], GVHD [44], and FN [61].

Allo-HSCT Complications and Treatments Based on Targeting the Gut Microbiota

Modulating the levels of gut microbiota metabolites can play a role in ameliorating post-transplant complications. Pretransplantation administration of IL-25 may improve GVHD by protecting goblet cells against gut bacterial translocation [168]. Glucagon-like peptide-2 (GLP-2) is an enteroendocrine tissue hormone produced by intestinal L cells. Treatment with GLP-2 agonists reduces new-onset aGVHD and steroid-refractory GVHD [169]. Oral administration of *Mycobacterium avium* subsp. fragilis increases SCFAs, IL-22, and regulatory T-cells in the intestinal tract, maintaining intestinal integrity and ameliorating GVHD [170]. It has been suggested that prophylactic antibiotics to reduce microbial abundance can prevent the onset of aGVHD [171]. However, the reliability of this therapy has not yet been demonstrated, as it was previously thought that a reduction in microbial abundance would lead to a decrease in the levels of anti-inflammatory SCFAs and an increase in the development of GVHD [172]. Dysbiosis of the intestinal flora has been associated with pulmonary complications after HSCT [173]. LPS induces lung injury through TLR-4 signaling, and TLR4 antagonist therapy may protect against transplantation-related lung injury after HSCT [174]. Lower respiratory tract infections have been found to be fivefold less likely to occur in HSCT patients with higher levels of SCFA-producing microbial communities [175].

FMT involves the transfer of fecal matter from a healthy donor into a recipient whose gastrointestinal microbiota equilibrium has been disturbed. Through the introduction of a healthy microbiota, the microbiota composition in the recipient is restored to a balanced state [176]. In steroid-resistant aGVHD patients who received FMT, symptoms were effectively relieved, and no adverse effects occurred [177]. In a prospective, single-center, single-arm study, complete clinical responses were observed within 1 month in 10 of 15 patients with GVHD who received FMT, and no other interventions were imposed, with successful tapering of immunosuppressive medication in 6 of them and improved long-term survival [178]. This shows the great potential of FMT in the treatment of steroid-resistant HSCT complications.

CONCLUSIONS

Treatment regimens such as chemotherapy and TBI administered before and after allo-HSCT can affect the body's GM composition and OS levels. Changes in the GM affect post-transplant hematopoietic function and immune reconstitution, increasing the risk of postoperative secondary infections. These changes are associated with the development and severity of GVHD, hematologic tumor recurrence, and CFS. High ROS often results in poor hematopoietic reconstitution, increases the risk of infection, and triggers the development and progression of PGF and PT. Local levels of OS promote an inflammatory response that exacerbates damage to GVHD target cells and exacerbates GVHD symptoms. OS in vascular ECs and tracheal epithelial cells is associated with allo-HSCT complications such as TA-TMA and BOS. GM surface receptors and metabolites not only affect intestinal OS, but can also enter the circulation through the intestinal barrier to reach the lesion and the bone marrow hematopoietic microenvironment through various signaling pathways. This affects OS levels; regulates hematopoietic differentiation, body immunity, and inflammatory responses; and is responsible for a variety of allo-HSCT complications and exacerbations. This review provides a new perspective for clinical prevention and treatment of allo-HSCT complications, with the hope that modulating OS through gut microbial therapies will achieve prevention and mitigation of allo-HSCT complications and improve post-transplant survival.

Conflict of Interest Disclosure

The authors do not have any conflicts of interest to declare in relation to this work.

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Data Availability

Further inquiries about data sets generated and/or analyzed can be directed to the corresponding authors.

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